Interpreting Parameter Estimates.

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HEALTH
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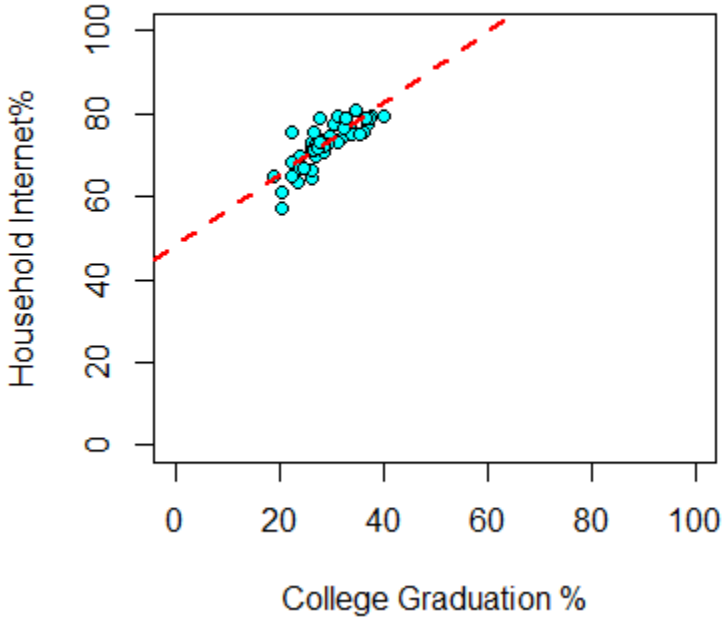
Interpreting Linear Parameters

```
> head(DAT1)
  State Internet College
1    AK     63.5    23.5
2    AL     79.0    28.0
3    AR     60.9    20.6
4    AZ     73.9    27.4
5    CA     77.9    31.0
6    CO     79.4    37.8
…
```



Slope

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = 0.86$$

Intercept:

$$b_0 = \bar{y} - b_1\bar{x} = 47.91$$

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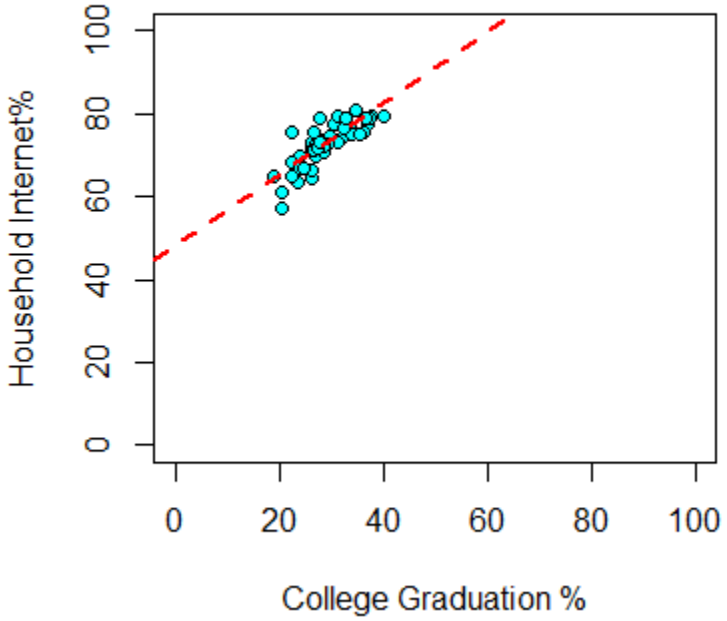
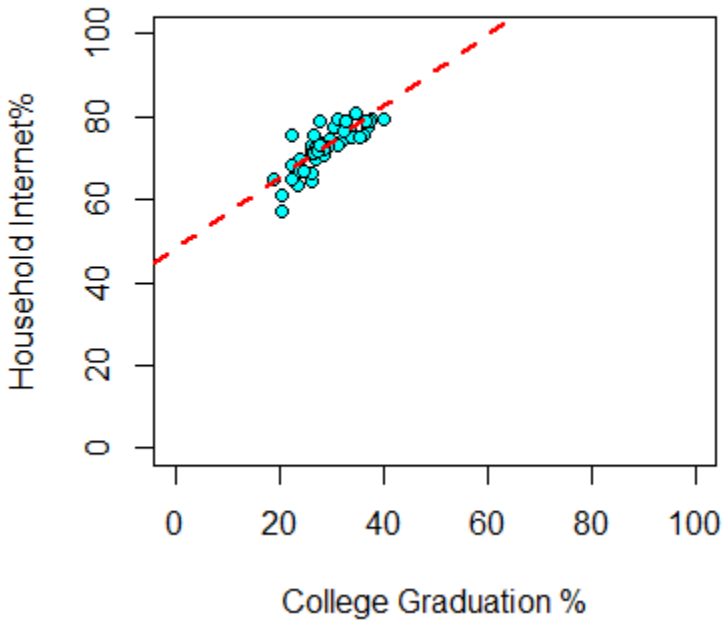


- The intercept is estimated the value of y when all x's are zero.
- The slope is the estimated change in y for a 1-unit change in x.
- These general rules will always be true. For more specific interpretations, it depends on exactly what x and y are (scale, type of variable, etc).

Untransformed Variables

```
lm(formula = Internet ~ College, data = DAT1)

Coefficients:
(Intercept)      College
    47.9123       0.8619
```



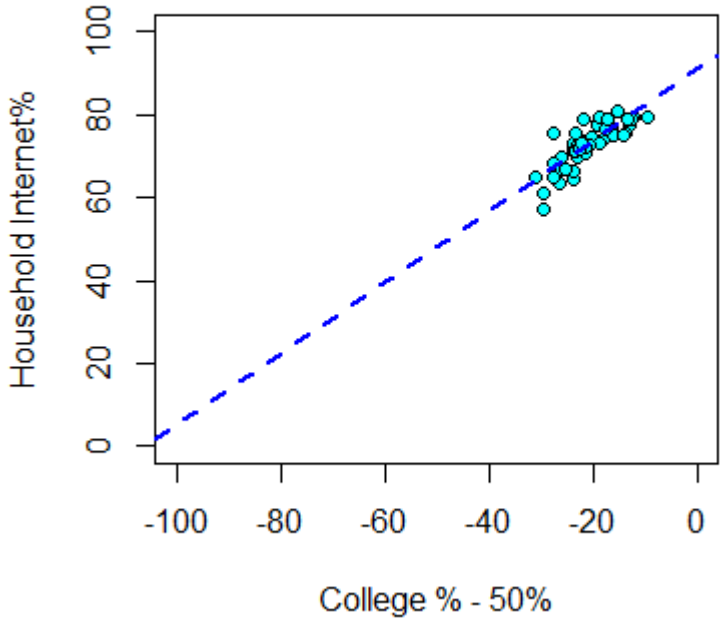
Value-Centered Variable

```
DAT1$College50<-DAT1$College-50
```

$x_{50i} = x_i - 50$

```
lm(formula = Internet ~ College50, data = DAT1)

Coefficients:
(Intercept)    College50
    91.0049       0.8619
```



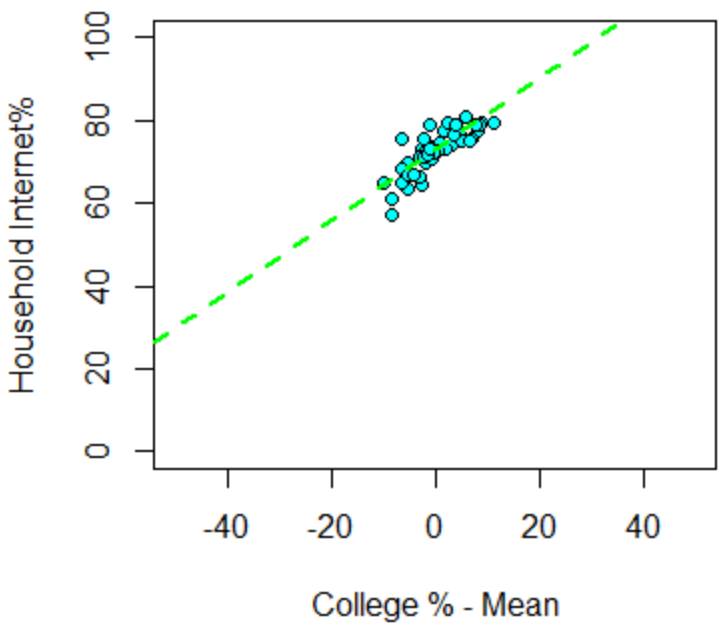
Mean-Centered Variable

```
DAT1$College.c<-DAT1$College-mean(DAT1$College)
```

$x_{ci} = x_i - \bar{x}$

```
lm(formula = Internet ~ College.c, data = DAT1)

Coefficients:
(Intercept)    College.c
    72.8060       0.8619
```



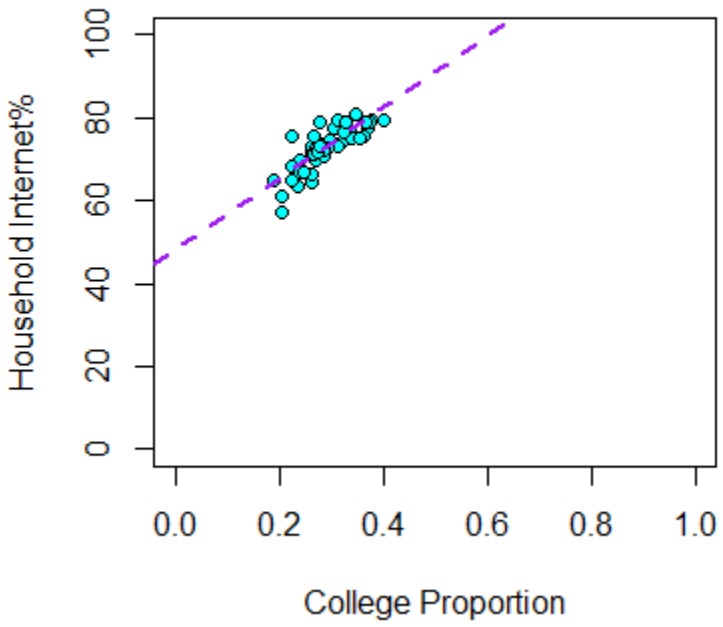
Changing the Scale of X

```
DAT1$College.p<-DAT1$College/100
```

$x_{pi} = x_i/100$

```
lm(formula = Internet ~ College.p, data = DAT1)

Coefficients:
(Intercept)    College.p
      47.91        86.19
```



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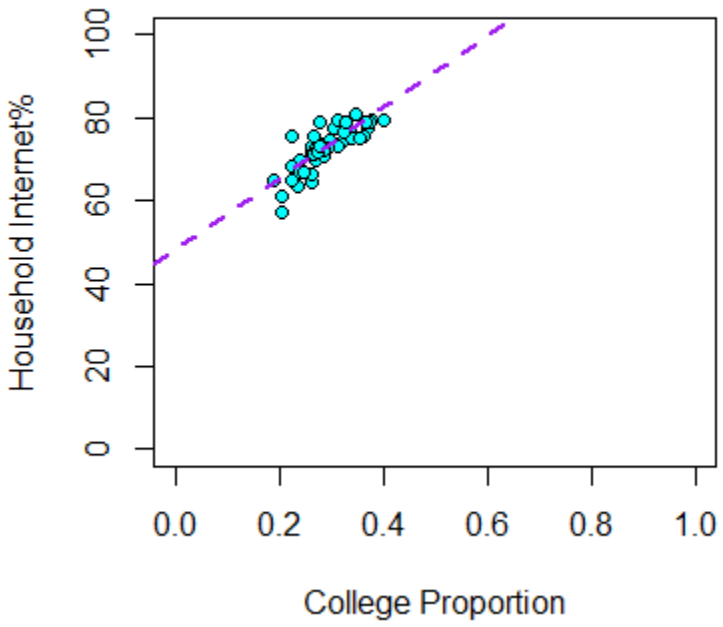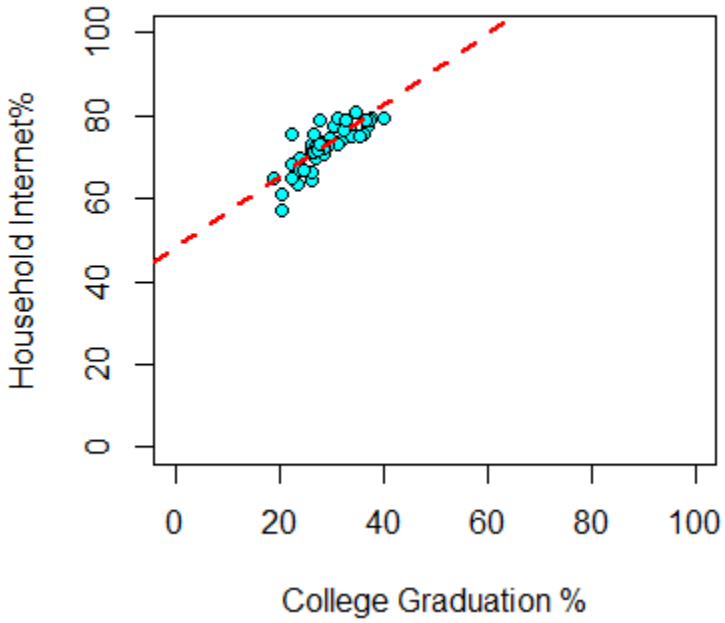
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